

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 - Scenario-Based Questions

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO4 – Scenario-Based Questions	Assessment:	Assessment Tool 1– Scenario-Based Questions
Weightage:	40% of CIA	Total Marks:	100
CO4 Statement:	Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)		
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Section / Batch:	CSE-DS	Date:	20-08-2026

Code of Conduct

I Abzam A (192573029) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Abzam A

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	

Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

QUESTIONS :

1. SOFTWARE APPLICATION

Topic: Hypothesis Testing Using p-value

A software company claims that its new application has an average response time of less than 2 seconds. A hypothesis test gives a p-value of 0.004.

At $\alpha = 0.01$, determine whether there is sufficient evidence to support the company's claim.

2. DIGITAL MARKETING

Topic: Hypothesis Testing Using p-value

A digital marketing team launches a new advertisement and wants to know whether it increases the click-through rate. Before the campaign, the click-through rate was 6%. After the campaign, a hypothesis test produces a p-value of 0.12.

What should the data science team conclude at the 5% significance level?

3. MACHINE LEARNING

Topic: Hypothesis Testing Using p-value

A machine-learning team develops a new classification model. The old model has an accuracy of 82%, while the new model achieves 85%. A statistical test gives a p-value of 0.01.

Explain what the p-value means and determine whether the team should conclude that the new model performs significantly better.

4. MANUFACTURING / QUALITY CONTROL

Topic: Confidence Interval (CI) + p-value

A factory claims that its defect rate is 2%. After randomly inspecting products, a 95% confidence interval for the defect rate is found to be (2.4%, 3.6%). At the same time, a hypothesis test gives a p-value of 0.01.

Interpret both results and explain whether the company's claim should be rejected.

5. ONLINE SHOPPING / E-COMMERCE

Topic: Confidence Interval (CI) + p-value

An online shopping company wants to determine whether a new user-interface design increases the average amount spent per customer. The data scientist calculates a 95% confidence interval for the difference in average spending as (₹120, ₹480) and obtains a p-value of 0.002.

Interpret the confidence interval and p-value, and provide a business recommendation.

ANSWERS:

1. SOFTWARE APPLICATION – HYPOTHESIS TESTING USING p-VALUE

Given Information

Company's claim: Average response time is less than 2 seconds.

$$p\text{-value} = 0.004$$

$$\text{Significance level, } \alpha = 0.01$$

1. State the Hypotheses

$H_0: \mu \geq 2$ seconds (or, for testing purposes, $H_0: \mu = 2$ seconds)

$$H_1: \mu < 2 \text{ seconds}$$

Here, μ represents the true average response time of the new application.

2. Compare the p-value with α

$$p\text{-value} = 0.004$$

$$\alpha = 0.01$$

Since $0.004 < 0.01$, reject H_0 .

3. Conclusion

There is sufficient statistical evidence at the 1% significance level to support the company's claim that the application's average response time is less than 2 seconds.

Industry-Oriented Conclusion

The software company can claim, based on the hypothesis test, that the new application has a statistically supported average response time below 2 seconds. This indicates good application performance and responsiveness.

2. DIGITAL MARKETING – HYPOTHESIS TESTING USING p-VALUE

Given Information

Before the campaign, click-through rate = 6%.

The aim is to determine whether the new advertisement increases the click-through rate.

$$p\text{-value} = 0.12$$

$$\text{Significance level, } \alpha = 0.05$$

1. State the Hypotheses

$$H_0: p \leq 0.06 \text{ (or, for testing purposes, } H_0: p = 0.06)$$

$$H_1: p > 0.06$$

Here, p represents the true click-through rate after the campaign.

2. Compare the p-value with α

$$p\text{-value} = 0.12$$

$$\alpha = 0.05$$

Since $0.12 > 0.05$, fail to reject H_0 .

3. Conclusion

There is not sufficient statistical evidence at the 5% significance level to conclude that the new advertisement increases the click-through rate.

Industry-Oriented Conclusion

The data science team should not conclude that the advertising campaign significantly improved the click-through rate. The team may collect more data, test the campaign further, or modify the advertisement before making a large-scale investment.

3. MACHINE LEARNING – HYPOTHESIS TESTING USING p-VALUE

Given Information

Old model accuracy = 82%

New model accuracy = 85%

p-value = 0.01

1. State the Hypotheses

H_0 : The new model does not perform better than the old model.

H_1 : The new model performs better than the old model.

2. Meaning of the p-value

A p-value of 0.01 means that, assuming the null hypothesis is true, the probability of obtaining a result at least as extreme as the observed improvement is about 1%.

3. Decision

At the common 5% significance level, $\alpha = 0.05$.

Since $0.01 < 0.05$, reject H_0 .

4. Conclusion

The improvement from 82% to 85% is statistically significant at the 5% significance level. Therefore, the team has sufficient evidence to conclude that the new model performs significantly better than the old model.

Industry-Oriented Conclusion

The machine-learning team should consider adopting or further validating the new model because the observed improvement in classification accuracy is statistically significant.

4. MANUFACTURING / QUALITY CONTROL – CI + p-VALUE

Given Information

Company's claimed defect rate = 2%

95% confidence interval for the defect rate = (2.4%, 3.6%)

p-value = 0.01

1. Interpret the Confidence Interval

The 95% confidence interval indicates that the true defect rate is estimated to lie between 2.4% and 3.6%.

The claimed defect rate of 2% is not contained in this interval.

2. Confidence Interval Decision

Since 2% is outside the 95% confidence interval, the company's claim of a 2% defect rate should be rejected at the 5% significance level.

3. Interpret the p-value

The p-value of 0.01 is less than 0.05.

Therefore, reject H_0 at the 5% significance level.

4. Compare Both Results

Both the confidence interval and p-value approaches lead to the same conclusion: reject the company's claim.

Industry-Oriented Conclusion

There is sufficient statistical evidence that the actual defect rate is different from, and appears higher than, 2%. The factory should investigate its production process and strengthen quality-control measures to reduce defects.

5. ONLINE SHOPPING – CI + p-VALUE

Given Information

95% confidence interval for the difference in average spending = (₹120, ₹480)

$$p\text{-value} = 0.002$$

1. Interpret the Confidence Interval

The confidence interval shows that the new user-interface design is estimated to increase average customer spending by between ₹120 and ₹480.

Because the entire interval is positive and does not include ₹0, the new design indicates a statistically significant increase in average spending.

2. Interpret the p-value

$$p\text{-value} = 0.002$$

$$\text{At } \alpha = 0.05, 0.002 < 0.05.$$

Therefore, reject H_0 of no increase in average customer spending.

3. Compare Both Results

Both the confidence interval and p-value provide the same conclusion: the new user-interface design significantly increases the average amount spent per customer.

4. Business Recommendation

The online shopping company should consider implementing the new user-interface design more widely because it is associated with a statistically significant increase in customer spending. Before a full rollout, the company may also evaluate implementation cost, user experience, and long-term performance.